



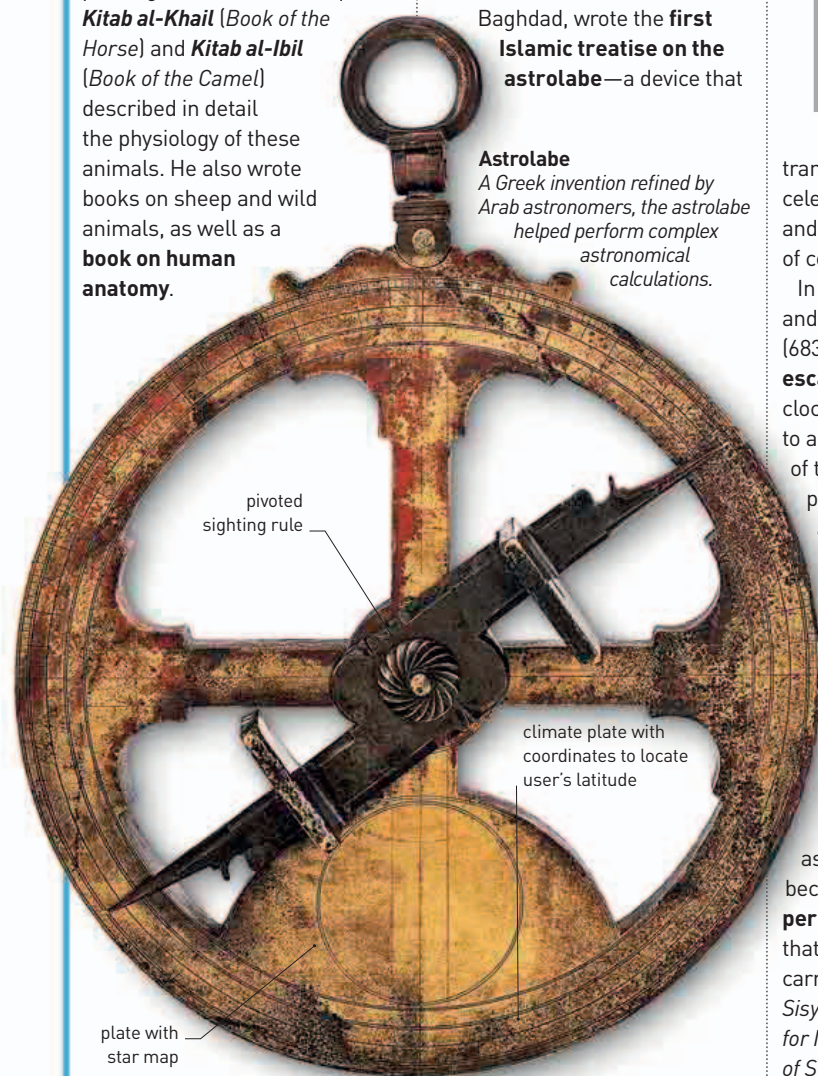
This image depicts Jabir ibn-Hayyan giving a lecture on alchemy in his home town, Edessa, modern Turkey. The town played an important role in the transmission of Greek science into the Islamic world.

**THE ISLAMIC WORLD'S FIRST MAJOR TREATISES** on zoology were produced by al-Asmai, a philologist from Basra, Iraq. His *Kitab al-Khail* (*Book of the Horse*) and *Kitab al-Ibil* (*Book of the Camel*) described in detail the physiology of these animals. He also wrote books on sheep and wild animals, as well as a **book on human anatomy**.

As knowledge of Greek astronomy spread to the Islamic world, Ibrahim al-Fazari (d. c.796), an astronomer from Baghdad, wrote the **first Islamic treatise on the astrolabe**—a device that

#### Astrolabe

A Greek invention refined by Arab astronomers, the astrolabe helped perform complex astronomical calculations.



## 13 THE NUMBER OF TEST SITES SET UP BY YI XING FOR HIS ASTRONOMICAL SURVEY

transferred observations of the celestial sphere onto flat plates and helped predict the location of celestial bodies.

In China, around 725, engineer and astronomer Yi Xing (683–727) invented the **first escapement** for a mechanical clock. The device was attached to an armillary sphere (a model of the celestial sphere) that was powered by water. It used a toothed gear to transfer energy to the moving parts of the sphere and to regulate their movement. Yi Xing also carried out a **major astronomical survey** to help predict solar eclipses more accurately and reform the calendar.

In India, mathematician and astrologer Lalla (c.720–790) became the first to **describe a perpetuum mobile**, a machine that once set in motion would carry on moving forever. His *Sisyadhivrdhdhitantra* (*Treatise for Increasing the Intelligence of Students*) also gave details

of **planetary movements, conjunctions, and eclipses**, although he rejected the idea that Earth rotated.

A few years later, in 762, the **city of Baghdad** was founded by the caliph al-Mansur. The **first planned city in the Islamic world**, its perfectly round shape was laid out by al-Naubakht, a Persian astrologer. His son, al-Fadl ibn Naubakht, founded the House of Wisdom in Baghdad, which became

a major Islamic center for the study of science.

**Jabir ibn-Hayyan** (c.722–804) was an early Islamic alchemist who has become known as the father of Arab chemistry. He invented the alembic, an enclosed flask for heating liquids, established the **classification of substances** into metals and nonmetals, and identified the **properties of acids and alkalis**.

Jurjish ibn Bakhtishu was the first of a dynasty of **Islamic physicians** who served the Abbasid caliphs at Baghdad. He rose to prominence when he cured the caliph al-Mansur of a stomach complaint in 765. His grandson Jibril **founded the first hospital in Baghdad** some time after 805.



#### ALCHEMY

First developed in Hellenistic Egypt (4th–1st century BCE) by scholars such as Zosimos of Panopolis, alchemy was advanced further by Arab practitioners such as ibn-Hayyan and al-Razi in the 8th–9th centuries. It was concerned mainly with the transmutation of base metals, such as lead, into noble metals, such as gold, through the use of the “philosopher’s stone.” It led to the development of many practical chemical processes, such as distillation and fermentation.

By 700 Lee boards and center boards are used on Chinese ships

c.700 Colored tuster-painted glass is created in the Islamic world

c.704 Anglo-Saxon monk Bede writes a treatise on timekeeping

c.720 Chinese exhibit knowledge of magnetic declination

c.725 Yi Xing invents escapement for mechanical clock

c.750 Al-Fazari writes a treatise on the astrolabe

c.750 Paper use spreads from China to the Islamic world

762 First planned city in the Islamic world—Baghdad—is built

c.750 Arabs establish cotton textile industry in Spain

771 Siddhanta, Indian astronomical treatise, is translated into Arabic

765 Jurjish ibn Bakhtishu rises to prominence after curing caliph al-Mansur of a stomach complaint

c.775 Ibn-Hayyan invents enclosed alembic flask for heating fluids

785 Imperial cartographer Jia Dan completes the first map of China

c.775 Indian mathematician Lalla describes perpetual motion device

c.800 Al-Asmai writes treatises on zoology